Applied Financial Math for Decision-Making

Five client decisions (annuities, fund returns, capital budgeting, bonds and money-market yields) solved, re-derived and published as an interactive dashboard.

Ajai Upadhyaya

ABSTRACT This report documents a course case study set at a fictional wealth manager, Dynamic Wealth Management, and the dashboard built to present it. Each of the five answers was recomputed here from the course workbook and matches the dashboard to rounding error. Plan C, a growing perpetuity worth $500,000, beats two fixed annuities. The OceanBlue fund's time-weighted return (9.16%) exceeds its money-weighted return (8.11%). Facility Type B has the highest NPV ($1.72M). Bond C pays the highest current yield (7.85%), and Investment C the highest effective annual yield (20.71%). The Case 1 winner depends on an input conflict in the course materials. At the $15,000 first payment the brief states, Plan A wins.

01

Summary

The project is a submission for a course on applied financial mathematics. The student plays a senior advisor at Dynamic Wealth Management, a fictional New York firm, and answers five client questions from a supplied Excel workbook. The work went beyond the workbook: a small TypeScript engine solves every case, a web dashboard presents the answers, and a four-page LaTeX memo is typeset from the same computed numbers.

$500,000	9.16%	$1.72M	7.85%	20.71%
Case 1 · PV of Plan C, the growing perpetuity	Case 2 · OceanBlue TWRR (MWRR 8.11%)	Case 3 · NPV of facility Type B at 12%	Case 4 · current yield of Bond C	Case 5 · effective annual yield of Investment C

KEY FINDINGS

1. **All five answers reproduce.** Every case was recomputed for this report directly from the workbook cells and matches the dashboard's stored results to within rounding error (10^{-8} relative or better).
2. **The Case 1 winner rests on a disputed input.** The brief gives Plan C a $15,000 first payment, the dashboard uses $20,000, and the workbook's cash-flow cells compound $20,000 at 3%. Plan C wins at $20,000 but loses to Plan A at $15,000. The break-even first payment is $15,340.
3. **The robust answers are robust.** Type B beats Type A at every positive discount rate, not just at 12%, and Type C has no IRR at all. Investment C ranks first on every yield measure.
4. **Some rankings are artefacts of the metric.** All three bonds are priced to the same 7% yield, so current yield ranks them by coupon alone. Holding-period yield ranks Investment B above A; annualising reverses that order.

Table 1. The five decisions.

Case	Client and question	Measure	Answer
1	Mr. Thompson: which of three annuity plans to take	Present value at 6%	Plan C, $500,000 (dashboard inputs)
2	Emma and David Brooks: how OceanBlue Fund performed	MWRR and TWRR	TWRR 9.16% exceeds MWRR 8.11%
3	James Mercer, Orion Technologies: which facility to build	NPV and IRR at 12%	Type B: NPV $1.72M, IRR 21.66%
4	Andrew Thompson: which bond has the highest current yield	Price and current yield at 7%	Bond C: $1,018.51, 7.85%
5	Lisa Moore: which money-market instrument to buy	HPY, EAY, MMY	Investment C: EAY 20.71%

Clients, funds and companies are all fictional, and every input comes from the course materials. Nothing here is investment advice. The value of the work is in how it is done: each answer comes from one engine, is checked against the workbook, and appears in the dashboard and the memo as the same number.

02

Case 1 · Valuing three annuities

Mr. Thompson has an inheritance and three plans to choose from. Plan A pays $30,000 a year for 25 years. Plan B pays $50,000 a year for 15 years, starting at the end of year 11. Plan C is a growing perpetuity whose payments rise 2% a year. The course fixes the discount rate at 6% and asks for present values.

$$PV_A = A \cdot [1 - (1 + r)^{-N}] / r \qquad PV_B = PV_{annuity}(A, 15, r) / (1 + r)^{10} \qquad PV_C = C / (r - g) \tag{1}$$

At 6%, Plan A is worth $383,500.68 and Plan B $271,163.46. Discounting the workbook's explicit year-by-year cash-flow rows gives the same two values, so the closed forms and the sheet agree. Plan B's larger payments do not make up for ten years of waiting: it is worth less than Plan A at every discount rate from 3% to 10%.

The Plan C input conflict

The course materials disagree about Plan C. The case text, and the description cell on the workbook's own Case 1 sheet (D4), give an initial payment of $15,000 growing 2%. The sheet's cash-flow row starts at $20,000 and grows each cell by a factor of 1.03 (E21:J21). The dashboard and memo use a third combination, $20,000 growing 2%, which matches neither source exactly. The dashboard states that the Excel sheet models $20,000 growing 2%, but the sheet's own formulas grow at 3%.

Table 2. Plan C under each reading of the inputs, at r = 6%. Plan A = $383,500.68; Plan B = $271,163.46.

Source of the Plan C inputs	First payment	Growth	PV of Plan C	Best plan
Case brief; workbook text cell D4	$15,000	2%	$375,000	**A**
Dashboard and memo (used for the published answer)	$20,000	2%	$500,000	**C**
Workbook cash-flow cells E21:J21	$20,000	3%	$666,667	**C**

Break-evens against Plan A at 6%: first payment $15,340.03 at 2% growth; growth 2.089% at a $15,000 first payment.

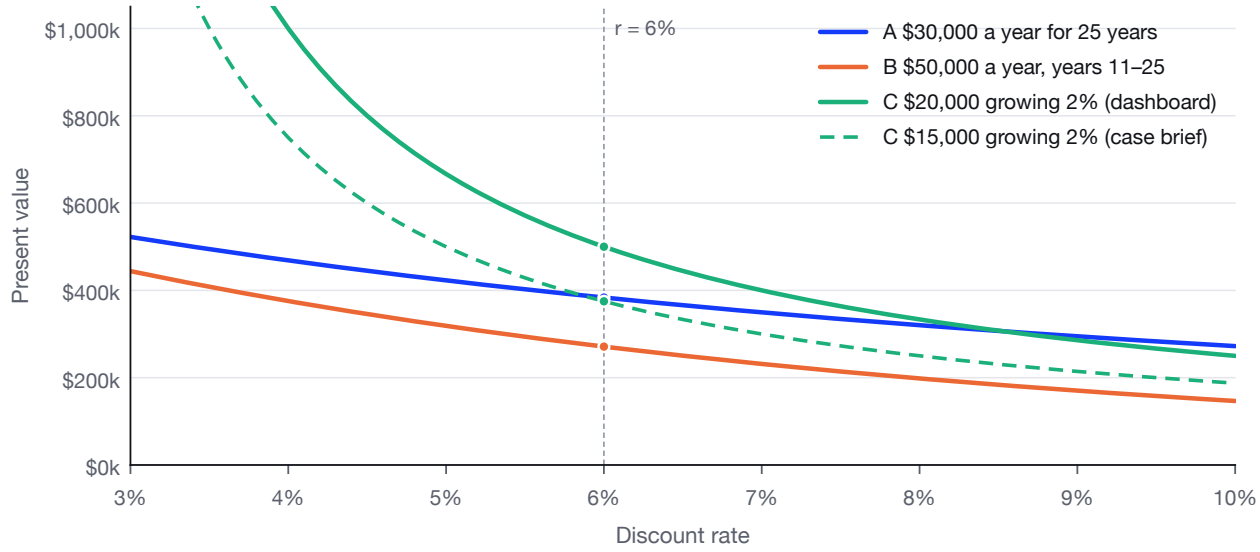


Figure 1. Present value against discount rate. Plan C at $20,000 (solid) beats Plan A for any rate below 8.53%. At the brief's $15,000 (dashed) it wins only below 5.86%, just under the course's 6%.

Source: case-study workbook, sheet "Case 1"; recomputed for this report (derived.json, case1).

The published answer is Plan C, and both the memo and the dashboard report the $15,000 alternative alongside it. The sensitivity shows how close the call is. A first payment just $340 above the brief's figure, or growth 0.09 points faster, would tie Plan C with Plan A. The recommendation depends on which reading of the inputs the grader takes, not on the valuation method.

03

Case 2 · Money-weighted versus time-weighted return

Emma and David Brooks are considering the OceanBlue Fund and want its past performance measured two ways. The workbook gives five years of assets under management and net returns. Investors are assumed to add or withdraw money at the start of each year, so each year's net flow is that year's opening AUM minus the previous year's closing value.

Table 3. OceanBlue Fund, $ millions. Net cash flow is from the investor's side: negative means money paid in.

Year	Opening AUM	Net return	Closing value	Net cash flow at start of year
1	75.00	+20%	90.00	−75.00
2	105.00	−5%	99.75	−15.00
3	55.00	+10%	60.50	+44.75
4	65.00	+20%	78.00	−4.50
5	85.00	+3%	87.55	−7.00
End				+87.55

Matches the course's sanity check: cash flows of −$75M and −$15M in the first two periods.

$$0 = \Sigma_t \, CF_t / (1 + MWRR)^t \qquad TWRR = [\Pi_t \, (1 + R_t)]^{1/5} - 1 \tag{2}$$

The time-weighted return is 9.16% a year. It links the five yearly returns geometrically (cumulative growth 55.0%) and is what the manager earned per dollar, whenever the dollar arrived. The money-weighted return, the internal rate of return on the investors' own cash flows, is 8.11%, which is 1.05 points lower. These cash flows change sign three times, so more than one IRR is possible in principle. A scan of rates from −90% to 1,000% finds only one root, so the 8.11% figure is unambiguous.

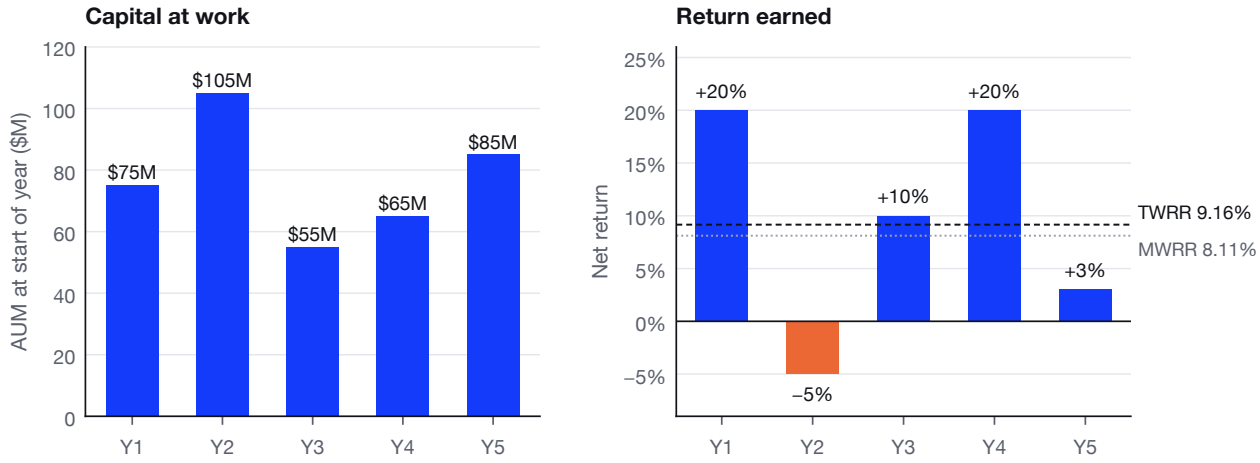


Figure 2. The fund held the most capital, $105M, in its one losing year (–5%) and the least, $55M, going into a +10% year. Money-weighting penalises that timing; time-weighting ignores it.

Source: case-study workbook, sheet "Case 2"; recomputed for this report (derived.json, case2).

The gap is about timing, not skill. The investors had their largest balance in the fund during year 2, its only loss, and then withdrew $44.75M before a +10% year. The TWRR suits judging the manager. The MWRR is closer to what these investors actually experienced. The memo answers the case's question directly: the time-weighted return is the higher of the two.

04

Case 3 · Choosing a semiconductor facility

Orion Technologies' CIO, James Mercer, is choosing between three facility designs with different capital outlays. The opportunity cost of capital is 12%. The cash flows come from the workbook.

Table 4. Facility cash flows ($M), NPV at 12% and IRR.

Type	t = 0	1	2	3	4	5	NPV at 12%	IRR
A	−9.0	4	3	3	3	1	1.572	20.01%
B	−7.0	2	3	3	2	2	1.719	21.66%
C	−4.3	10	−6	−2	—	—	−1.578	none

$$NPV(k) = \Sigma_t \, CF_t / (1 + k)^t \qquad NPV(IRR) = 0 \tag{3}$$

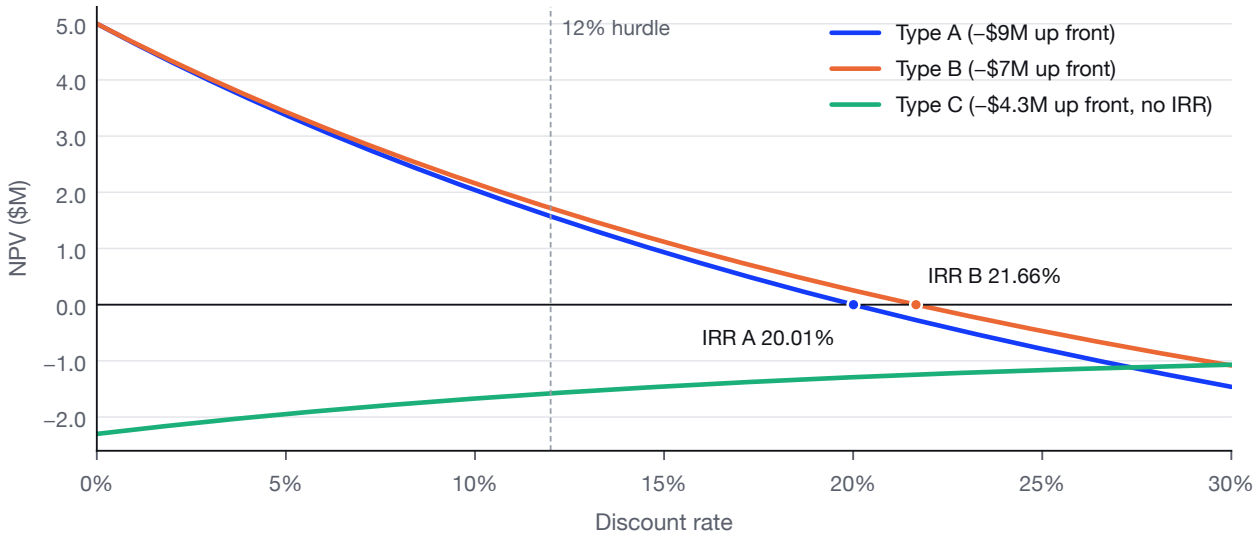


Figure 3. NPV profiles. Types A and B return the same undiscounted $5M, so their profiles meet at 0%. Above it, B leads at every rate. Type C never reaches zero.

Source: case-study workbook, sheet "Case 3"; recomputed for this report (derived.json, case3).

The recommendation is Type B, and it is robust. B has the highest NPV and the highest IRR, so the two rules agree. Its NPV exceeds A's at every positive discount rate tested, from 0.05% to 100%, so the choice does not depend on the 12% hurdle. B also needs $2M less up front: per dollar invested it creates $0.25 of NPV, against A's $0.17. Type C should be rejected. Its cash flows change sign twice and have no IRR. Its NPV is negative at every discount rate, peaking at −$0.88M near 58%. The dashboard marks its IRR "N/A", which is correct and not a solver failure. One of its labels, "Non-unique / N/A", overstates the case: there is not more than one IRR, there is none.

05

Case 4 · Three bonds at a 7% market rate

Andrew Thompson wants fixed income and is choosing between three $1,000 bonds. Bond A pays a 6% annual coupon for four years, Bond B 4% semiannually for three years, and Bond C 8% quarterly for two years. Each is priced by discounting its coupons and face value at the 7% market rate, divided by the coupon frequency m. Current yield is the annual coupon divided by price.

$$P = \Sigma_{i=1\ldots nm}\ (c{\cdot}F/m)\ /\ (1 + y/m)^{i} + F\ /\ (1 + y/m)^{nm} \qquad CY = c{\cdot}F\ /\ P \tag{4}$$

Table 5. Prices and yields at a 7% nominal market rate.

Bond	Coupon	Payments	Price	Trades at	Current yield	Effective annual yield
A	6%, annual	4 × $60	$966.13	discount	6.21%	7.00%
B	4%, semiannual	6 × $20	$920.07	discount	4.35%	7.12%
C	8%, quarterly	8 × $20	$1,018.51	premium	7.85%	7.19%

Effective annual yield $(1 + 0.07/m)^{m} - 1$ is computed for this report; the case asks only for price and current yield.

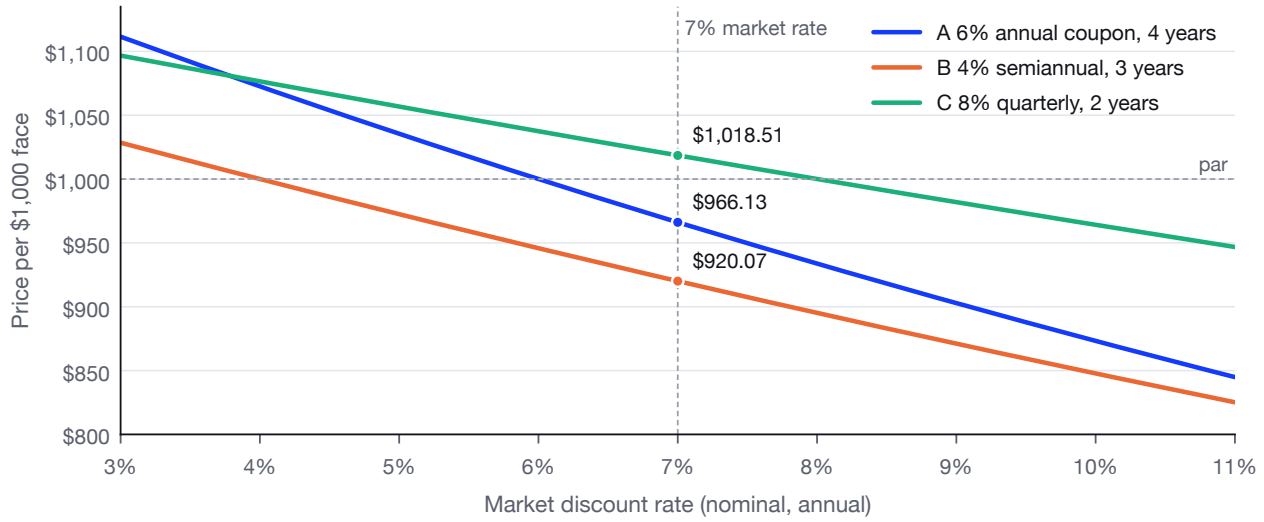


Figure 4. Price–yield curves. Each bond trades at par where the market rate equals its coupon rate. At 7%, only Bond C, with an 8% coupon, sits above par. Bond A's longer maturity gives it the steepest curve.

Source: case-study workbook, sheet "Case 4"; recomputed for this report (derived.json, case4).

Bond C has the highest current yield at 7.85%, the answer the case asks for. The result needs one qualification. All three bonds are priced off the same 7% nominal rate, so held to maturity each earns that same 7% nominal yield. Current yield ignores the pull to par, so it ranks them by coupon alone. C's high coupon comes with a $18.51 premium that is lost by maturity. B's low coupon comes with a $79.93 discount that is earned back. On an effective annual basis, C's quarterly compounding is worth slightly more (7.19%) than A's annual payment (7.00%).

06

Case 5 · Money-market yields

Lisa Moore is choosing between three $1,000 discount instruments with different prices and maturities. The case asks for three yield measures.

$$HPY = (P_1 - P_0) / P_0 \qquad EAY = (1 + HPY)^{365/t} - 1 \qquad MMY = HPY \times 360 / t \tag{5}$$

Table 6. Holding-period, effective annual and money-market yields.

Investment	Price	Days to maturity	HPY	EAY	MMY
A	$960	90	4.17%	18.00%	16.67%
B	$950	140	5.26%	14.31%	13.53%
C	$940	120	6.38%	20.71%	19.15%

Investment C ranks first on every measure, with an effective annual yield of 20.71%. A and B are where the choice of measure matters. B's holding-period yield (5.26%) beats A's (4.17%), but it takes 140 days to earn against A's 90. Once annualised, the order flips: A yields 18.00% and B 14.31% on an EAY basis. EAY compounds over a 365-day year and MMY scales simply over a 360-day year, so the two annualised measures differ in level, but they give the same ranking. Only an annualised measure compares instruments of different maturities fairly.

07

The dashboard

The answers are published through a small monorepo, with a web application and a quantitative library side by side.

- **Quant engine.** Pure TypeScript functions for annuities, return measurement, capital budgeting, bond pricing and money-market yields, with Newton iteration for IRR. The web app carries a mirrored copy so that its bundle is self-contained.
- **Pipeline.** An ingest script fingerprints the course workbook and writes its inputs to JSON. A verification script recomputes all five cases and writes them to results.json and to LaTeX macros, so the memo is typeset from the computed values instead of retyped ones. A third script pulls a macroeconomic snapshot from FRED: 10- and 2-year Treasury yields, the federal funds rate, CPI, unemployment and the BBB corporate yield. That data is context only and never enters a calculation.
- **Interface.** A Next.js 15 application (React 19, Tailwind 4, Recharts) with pages for an overview, each case, the quant library, macro and policy context, the memo, methods and the submission package. Case 1 has its own panel, which shows the $15,000 and $20,000 readings of Plan C side by side.
- **Build.** Continuous integration installs dependencies, runs ingest, verification and the macro fetch, then builds the site on every push.

The dashboard repeats the workbook's numbers without adding anything to them. There are no sliders for the discount rate or growth assumptions, and no automated tests. The verification script re-implements the formulas and reports what it computes, but asserts nothing. The ingest step does not read the workbook's cells either: it records the file's hash and writes inputs transcribed by hand. Those transcriptions match the sheets for Cases 2–5, and for Case 1 they follow the reading in Table 2. There is one small display error: the overview page computes the 2s10s spread in percentage points but labels it in basis points.

08

Limitations

- **Textbook inputs.** All five cases use inputs supplied by the course. Clients, OceanBlue Fund and Orion Technologies are fictional, and no real fund, issuer or portfolio data is involved.
- **Case 1 depends on an input choice.** Three conflicting sources specify Plan C, and the published winner holds under two of them. The dashboard's note describes the workbook as growing at 2%, but the sheet's cells grow at 3%.
- **Narrow metrics, by design.** Case 4 asks for current yield, which ignores the gain or loss to par, and its prices use a nominal per-period rate (7%/m). Case 5's EAY and MMY use different day-count bases. Each answer is correct for the question asked, but none of them is a full investment comparison.
- **Verification without tests.** The engine has no unit tests. Agreement with the workbook rests on hand-transcribed inputs and on the independent recomputation in this report, which reproduced every published figure.
- **A frozen snapshot.** The macro data is a May 2026 snapshot, and the repository's whole history falls on a single day, 22 May 2026.

A

Data and reproducibility

REPOSITORY	ajaiupadhyaya/case-study, commit 0c0e13c, 22 May 2026. A monorepo with apps/web (the dashboard), packages/quant (the engine), scripts/ and memo/.
INPUTS	“Dynamic Wealth Management (DWM).xlsx”, five sheets “Case 1”–“Case 5” (SHA-256 prefix cd5b7a8979ee9894, matching the dashboard's ingest record). Case prompts are in case1.md–case4.md; case4.md holds both Case 4 and Case 5.
RESULTS FILE	data/processed/results.json, written by scripts/verify-cases.mjs; it also generates memo/generated/results.tex for the four-page memo.
THIS REPORT	Every case was recomputed in Python from the workbook cells and asserted equal to results.json. Derived here: Plan C variants and break-evens, the PV and NPV profiles, the IRR-uniqueness scan, the Type C maximum NPV, NPV per dollar invested, effective annual bond yields and the yield rankings. All four figures come from these computations.

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